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Research Article

A quasi-experimental study on the energy expenditure, exercise intensity, and rating of perceived exertion of a male bodybuilding posing training

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Abstract

This study aimed to evaluate the intensity of posing training in male bodybuilders by comparing it to vigorous intensity parameters and examining the effects of stimulant usage and preparation phases. Specifically, this study compared posing training to established vigorous intensity benchmarks using metabolic equivalents (METs) and heart rate (HR) responses and assessed differences between athletes using stimulants versus those not using stimulants, as well as during peak week versus other preparation phases. Fifteen male bodybuilding athletes (mean age: 32.07 ± 7.82 years; mean body mass: 92.89 ± 9.06 kg; mean height: 1.76 ± 0.05 m; mean body mass index: 29.78 ± 2.24 kg/m^2) completed four compulsory posing sets. Findings demonstrated that posing training can be classified as vigorous training intensity using METs (mean difference of −0.50 METs, $p = 0.067$, ES = −0.51) and maximum HR (mean difference of 14.55 bpm, $p = 0.009$, ES = 0.79) compared to the established values of 6.0 METs and 77% vigorous intensity of %HRmax, respectively. Additionally, athletes using stimulants exhibited higher ratings of perceived exertion (RPE) of 2.20 arbitrary units ($p = 0.008$) and maximum HR (mean difference of 24.37 bpm, $p = 0.005$, ES = 0.79) compared to those not using stimulants. During peak week, athletes showed higher RPE of 2.38 arbitrary units ($p = 0.004$) and maximum HR (mean difference of 14.55 bpm, $p = 0.009$, ES = 0.79) compared to other preparation phases. These results indicate that bodybuilding posing training meets the criteria for vigorous exercise intensity and that stimulant use and peak week significantly affect physiological responses and perceived exertion.

Novelty

- This study is novel in classifying bodybuilding posing training as vigorous intensity exercise using metabolic equivalents (METs) and heart rate (HR) responses. It provides empirical evidence showing that posing training meets the vigorous intensity benchmarks, with METs and %HRmax values comparable to established vigorous exercise standards.
- The research highlights the novel finding that stimulant usage and the peak week phase of preparation significantly influence physiological responses and perceived exertion in bodybuilders. Specifically, athletes using stimulants and those in peak week displayed higher ratings of perceived exertion (RPE) and maximum heart rates, indicating that these factors notably affect the intensity and perceived difficulty of posing training.

Key words: posing training, bodybuilding, athletes, aerobic training, resistance training

Résumé

Cette étude visait à évaluer l'intensité de l'entraînement de pose chez les culturistes masculins en le comparant aux paramètres d'intensité vigoureuse et en examinant les effets de l'utilisation de stimulants et des phases de préparation. Plus précisément, cette étude a comparé l'entraînement de pose aux repères établis d'intensité vigoureuse en utilisant les équivalents métaboliques (MET) et les réponses de la fréquence cardiaque (FC), et a évalué les différences entre les athlètes utilisant des stimulants et ceux n'en utilisant pas, ainsi qu'entre la semaine de pointe et les autres phases de préparation. Quinze athlètes masculins de culturisme (âge moyen: 32, 07 ± 7, 82 ans; masse corporelle moyenne: 92, 89 ± 9, 06 kg; taille moyenne:

1, 76 ± 0, 05 m; IMC moyen: 29, 78 ± 2, 24 kg/m^2) ont réalisé quatre séries de poses obligatoires. Les résultats ont démontré que l'entraînement de pose peut être classé comme une intensité d'entraînement vigoureuse en utilisant les MET (différence moyenne de −0, 50 MET, $p = 0, 067$, ES = −0, 51) et la FC maximale (différence moyenne de 14, 55 bpm, $p = 0, 009$, ES = 0, 79) par rapport aux valeurs établies de 6, 0 MET et 77% d'intensité vigoureuse de %HRmax, respectivement. De plus, les athlètes utilisant des stimulants ont montré des scores d'effort perçu (RPE) plus élevés de 2, 20 unités arbitraires ($p = 0, 008$) et une FC maximale (différence moyenne de 24, 37 bpm, $p = 0, 005$, ES = 0, 79) par rapport à ceux n'utilisant pas de stimulants. Pendant la semaine de pointe, les athlètes ont montré des RPE plus élevés de 2, 38 unités arbitraires ($p = 0, 004$) et une FC maximale (différence moyenne de 14, 55 bpm, $p = 0, 009$, ES = 0, 79) par rapport aux autres phases de préparation. Ces résultats indiquent que l'entraînement de pose en culturisme répond aux critères d'intensité d'exercice vigoureuse et que l'utilisation de stimulants et la semaine de pointe affectent significativement les réponses physiologiques et l'effort perçu.

Nouveauté

- Cette étude est novatrice en classant l'entraînement de posing en bodybuilding comme un exercice d'intensité vigoureuse en utilisant les équivalents métaboliques (MET) et les réponses de la fréquence cardiaque (FC). Elle fournit des preuves empiriques montrant que l'entraînement de posing atteint les critères de l'intensité vigoureuse, avec des valeurs de MET et de %HRmax comparables aux normes établies pour les exercices vigoureux.
- La recherche met en évidence la découverte novatrice que l'utilisation de stimulants et la phase de préparation de la semaine de pointe influencent significativement les réponses physiologiques et la perception de l'effort chez les bodybuilders. Plus précisément, les athlètes utilisant des stimulants et ceux en semaine de pointe ont affiché des niveaux plus élevés de perception de l'effort (RPE) et des fréquences cardiaques maximales, indiquant que ces facteurs affectent notablement l'intensité et la difficulté perçue de l'entraînement de posing.

Mots-clés : entraînement de pose, culturisme, athlètes, entraînement aérobique, entraînement en résistance

Introduction

Bodybuilding has gained prominence and popularity among resistance training (RT) practitioners and the overall public over the years (Castro et al. 2011). Bodybuilding is a sport in which competitors are evaluated on their muscle mass, proportion, and body definition, demonstrated through a variety of different poses (Helms et al. 2014).

In bodybuilding, athletes seek well-defined body shapes with aesthetically pleasing muscles, in addition to balanced symmetry between the different segments of the body according to each competition category (Almeida et al. 2023). Competitors perform a series of poses on stage, and are evaluated by judges based on aesthetic appearance considering the muscle mass, symmetry, and definition of each competitor (Cyrino et al. 2008; Hackett et al. 2013). Athletes who present higher scores in a greater number of poses prevail (Cyrino et al. 2008). Furthermore, the level of hypertrophy and muscle definition varies according to the category. For example, men who compete in classic bodybuilding have smaller muscle volume and extreme muscle definition, while senior bodybuilders have greater muscle volume along with definition (de Moraes et al. 2019).

Reducing subcutaneous fat is a critical component of achieving the desired physical appearance in bodybuilding (Antonio 2000). Throughout the preparation phase, bodybuilders routinely practice their posing routines to enhance muscle definition and symmetry, which are essential criteria for competition judging. Regular posing practice not only improves muscle conditioning and control but also enables athletes to present their physiques more effectively. While the primary purpose of posing is not to increase energy expenditure, it can contribute to caloric burn, which is beneficial during periods of fat loss (Kind and Helms 2023). This additional caloric expenditure aids athletes in managing their energy balance and optimizing their body composition for competition.

The difficulty of posing training is related to a high level of physical effort needed in the posing sessions, that is accompanied by vigorous muscle contractions (Alves et al. 2020). Also, stimulants such as clenbuterol and liothyronine that are commonly taken during peak-week (the week prior to the contest) in non-drug-free competitors might also affect posing training performance (Almeida et al. 2023). A previous study assessed mood states and gastrointestinal symptoms following carbohydrate loading, and authors suggested that a high prevalence of constipation and diarrhea might have hindered improvement in mood states (de Moraes et al. 2019). However, considering that abdominal pain is associated with constipation (Wong et al. 2020), these characteristics might alter performance during bodybuilding posing, and should not be overlooked.

During the preparation phase, particularly in the peak week, non-drug-free bodybuilders extensively use pharmacological aids such as testosterone propionate and clenbuterol (Almeida et al. 2023). In animal studies, clenbuterol has been shown to cause a significant increase in passive Ca^{2+} leak from the sarcoplasmic reticulum (SR) in both fast-twitch and slow-twitch muscle fibers (Bakker et al. 1998). This increase in Ca^{2+} leak reduces SR Ca^{2+} loading, thereby decreasing the amount of Ca^{2+} available for muscle contraction (Bakker et al. 1998). Furthermore, chronic clenbuterol treatment significantly reduces the ability of rats to exercise at high intensity (Bakker et al. 1998). This suggests that while clenbuterol can induce muscle hypertrophy, it might also impair muscle function and reduce overall exercise performance under certain conditions (Bakker et al. 1998; Kline et al. 2007).

Previous studies used different methods to define the energy expenditure during activities and different methods

to control exercise intensity, such as metabolic equivalents (METs), heart rate (HR), heart rate reserve, and rating of perceived exertion (RPE) (Ainsworth et al. 2000; Achten and Jeukendrup 2003; Robertson et al. 2003; Garber et al. 2011).

Energy expenditure in humans is influenced by size, body composition, environment, and behavior. A larger body, especially with a greater amount of fat-free mass, requires greater maintenance and therefore results in a higher basal metabolic rate or greater resting energy expenditure (Westerterp 2000). Moreover, the predominant energy systems for each type of exercise, whether through anaerobic or aerobic pathways, can vary in relation to mechanisms of action, energy sources, duration and performance (Shaner et al. 2014; Parks et al. 2020; Wilke et al. 2020, 2021). Thus, an issue in bodybuilding is the absence of physiologically based mechanisms to balance the indicators of external stimuli and adaptation processes. Elucidating these mechanisms will enable a significant improvement in the bodybuilding development process.

Maximal oxygen consumption kinetics have been studied in several types of exercise (Jones and Poole 2013). Regarding RT, a previous study found no significant differences in VO_2 between traditional RT sessions versus local muscular endurance RT (Almeida et al. 2011). Understanding the adjustments of metabolism and the cardiovascular system to posing can allow trainers to safely prescribe pose training as an alternative to exercise training to maximize energy expenditure and to improve dietetic calculations, particularly during periods where caloric expenditure needs to be closely monitored.

Based on these observations, the objective of the present study was to evaluate energy expenditure in a session of bodybuilding poses, RPE, hear rate response, and compare the intensity achieved during posing training to established vigorous intensity parameters reported in the literature, specifically using METs (6.0 METs) and percentage of maximum HR (%HRmax) (77% of estimated HRmax) (Garber et al. 2011). Additionally, the study sought to investigate differences in energy expenditure, RPE, and HR response between different phases of preparation, as well as between competitors using stimulants and those not using stimulants. We hypothesized that bodybuilding poses significantly elevate energy expenditure, accompanied by higher RPE and HR response. Additionally, we anticipated that athletes in peak week and those using stimulants would display higher RPE scores, mean HR, maximum HR, energy expenditure, and oxygen consumption compared to other phases of preparation and athletes not using stimulants.

Methods

Participants

This is a non-randomized controlled trial, and the report followed the Transparent Reporting of Evaluations with Non-randomized Designs (Des Jarlais et al. 2004). Fifteen male bodybuilders aged 32.07 ± 7.84 years, body mass of 92.89 ± 9.06 kg, height 1.76 ± 0.05, body mass index (BMI) of 29.78 ± 2.24 kg/m^2, body fat of 7.00 ± 3.55%, body fat of 6.43 ± 3.55 kg, and muscle mass of 82.92 ± 7.51 kg participated in the present study.

The inclusion criteria and exclusion criteria used were similar to a previous study and included competitive male bodybuilders of any subdivisions that provided reports of diet, nonsteroidal agents use, and anabolic androgenous substance (AAS) use (Almeida et al. 2023). Subjects were recruited through in-person conversations (referral and self-selection). Study was conducted between 21 May 2021 and 19 August 2022 at the Exercise Physiology Laboratory of the institutional research.

This study was approved by the local University Institutional Review Board (protocol number: 23163019.6.0000.0029) and was conducted in accordance with the Declaration of Helsinki (World Medical 2013). Subjects were informed about the study procedures and voluntarily signed an informed consent prior to enrollment. They were amateur bodybuilding athletes who had been competing frequently for more than 3 years. Athletes were in different stages of preparation, all were using AAS, and only a few were using stimulants ($n = 8$) such as caffeine and clenbuterol. All participants were evaluated at the same time of day (9:00–12:00 AM) and were fed 2 h prior the evaluation.

For power analysis, the study had a limited resource for the choice of the sample size to collect (Lakens 2022). Also, enrolling bodybuilders in this study was difficult. To obtain a power of 80% based on METs achieved in the posing time of 5.49 METs compared to 6.0 METs (vigorous intensity) (Garber et al. 2011), it would be necessary 31 participants. Thus, this is a low power study.

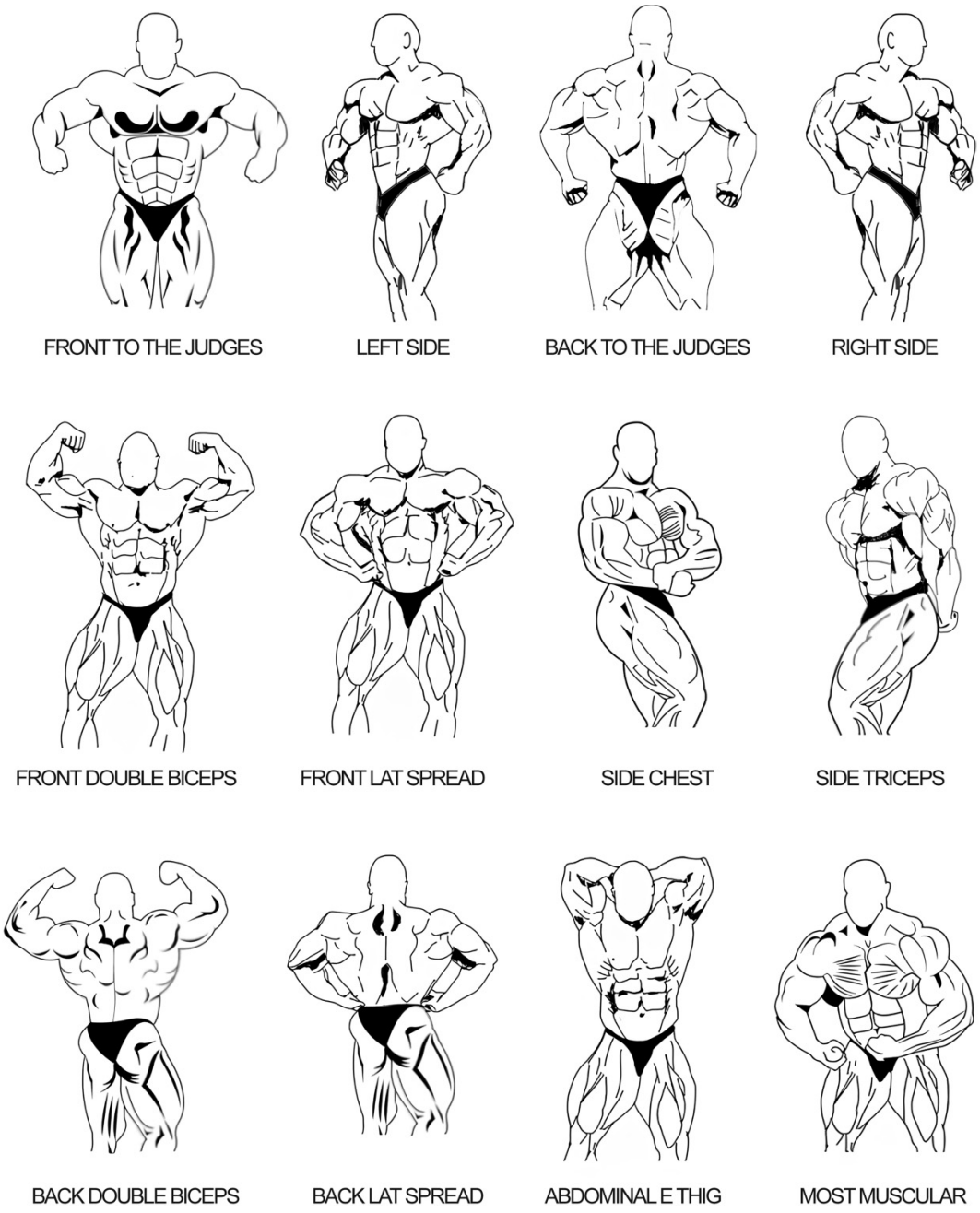
Interventions

Subjects arrived at the Laboratory of Exercise and Physiology and performed DXA analysis at first. Then, a bodybuilding pose familiarization routine was carried out and the participant remained for 10 s in each pose in the following sequence: front semi-relaxed, left profile, back, right profile, front double biceps, front expansion of lats, pectorals on the better side, back double biceps, back expansion of the lats, triceps on the better side, abdominals and thighs, and the most muscular positioning (Fig. 1). After that, a bodybuilding posing test following the same sequence was performed with RPE, HR, and metabolic measurements.

Body composition measurements

All subjects were weighed on a mechanical scale (Filizola, Sao Paulo, Brazil), and height was measured with a stadiometer built into the scale (precision: 0.5 cm) (Filizola, Sao Paulo, Brazil). Body mass index (BMI) was then calculated (body mass/height^2). After that, each participant underwent a supine whole body composition by dual-energy x-ray absorptiometry (DXA; Lunar Corp. Madson, WI USA, DPX-IQ 2446), performed by an experienced technically trained professional. DXA calibration was performed daily using a phantom. Body composition assessment by DXA in male bodybuilders display a large limit of agreement and caution should be made when interpreting body composition measurements in this group (Graybeal et al. 2020).

Fig. 1. Bodybuilding posing training.



Bodybuilding posing routine test

After familiarization, a bodybuilding posing sequence test was applied following the same order of the familiarization session. Subjects were advised to remain 10 s in each pose, and at the end of the entire sets of poses, a 45 s rest interval (RI) was taken to perform another set of poses, until completing a total of four sets.

Rating of perceived exertion

After each final set during RI, subjects were asked to report their RPE using the OMNI-RES (Robertson et al. 2003). The following instruction was adapted for the present study.

Instructions: We would like you to use these pictures to describe how your body feels during posing routine (show the OMNI-RES). You are going to perform a bodybuilding posing sequence, and after finishing each set, look at the person at the bottom, and at the top of the scale. Thus, if you feel somewhere in between Extremely Easy (0) and Extremely Hard (10), then give a number between 0 and 10. Please give a number that describes how your whole body feels.

Heart rate measurements and metabolic variables

Mean HR and maximum HR during the bodybuilding posing sequence were accurately recorded using a Polar chest

strap and Polar watch (RS800; Polar, USA). The device records HR continuously in real-time. The device captures heart HR data every second, providing detailed and accurate measurements. HR data collected during the 45 s RIs were excluded from the analysis. In addition, metabolic variables as expiratory exchange ratio, mean oxygen consumption (VO_2), and metabolic equivalent (MET) were measured during bodybuilding posing by the stationary calibrated device measuring in the breath-by-breath mode (Metalyzer 3B, CPX System, Cortex, Germany). To calculate the kilocalorie energy expenditure or total energy expenditure, the equation kilocalories = MET × weight × duration in hours, which was used in a previous study, was applied (Ainsworth et al. 2011).

Mean oxygen consumption ($VO_{2)}$ data collected during the 45 s RIs were excluded from the analysis. The mean values of VO_2 during bodybuilding posing were calculated and presented over specified time intervals, such as minute-by-minute averages, to provide a more stable and interpretable measure of oxygen consumption. Estimated maximum HR was also obtained by a previous equation to indirectly measure bodybuilding posing intensity (Tanaka et al. 2001).

Statistical analysis

Data are displayed by mean and standard deviation (SD), unless otherwise stated. Normality of data was verified by Shapiro–Wilk test. Outliers were presented only for % body fat, and fat mass (kg), respectively. Also, normality violation was present for % body fat, fat mass (kg), sets 3 and 4 of bodybuilding posing, and set 1 for RPE. For the time spent during posing, and RPE, the generalized mixed model (GMM) statistic was applied. Stimulants (e.g., clenbuterol use) and the phase of preparation (e.g., some participants were in peak-week) were used as covariates. The GMM was used instead of repeated measures Analysis of variance (ANOVA) because assumptions of homoscedasticity and sphericity when violated, results in an F-ratio that is inflated (Gueorguieva and Krystal 2004; de Melo et al. 2022).

Furthermore, GMM provides more robust and resilient models that reduce types I and II error rates (de Melo et al. 2022). The coefficient beta values were used as effect size in this study, and values of < 0.2–1.9, 0.2–0.4 and > 0.5 were considered as small, medium, and large, respectively (Fey et al. 2023). The *Hurvich and Tsai's criterion* (AICC) statistic for small sample sizes was used to evaluate the model fit (Field 2024). To verify the assumption and use of GMM model, diagnosis using residuals to identify outliers or potentially influencing observations was applied. Furthermore, the covariance structure used for the analyzes was *diagonal* as is the default covariance structure for repeated measures design (Field 2024).

To compare the exercise intensity of bodybuilding posing, as measured by METs, with the classification of exercise intensity for cardiorespiratory endurance exercise, a one-sample *t* test was performed. A one-sample *t* test is a statistical test used to determine whether the mean of a single sample is significantly different from a known or hypothesized population mean. In this study, the one-sample *t* test compares the mean METs value achieved during posing with an absolute intensity of 6.0 METs, which corresponds to a vigorous intensity level as defined in a previous study (Garber et al. 2011). Also, a comparison of mean and maximum HR values during bodybuilding posing with 77% of %HRmax (represented in this study by 142.45 bpm as vigorous intensity based on %HRmax) was applied (Garber et al. 2011). For variables measured at only one time point, an independent *t* test between participants (use of stimulants vs. no use of stimulants, and subjects at peak-week vs. other phases of preparation) was applied to verify possible differences. For *t* tests, Hedges' g effect size of 0.2, 0.5, and 0.8 were considered small, medium, and large, respectively (Lakens 2013; Serdar et al. 2021). The significance level adopted in this study was $p \leq 0.05$ and the software used for analysis was SPSS 20.0 and G*Power 3.1 (Faul et al. 2007).

Results

Table 1 displays the presentation time spent during bodybuilding posing, RPE, HR, and metabolic measurements. After comparing the METs with a vigorous intensity (6.0 METs), no statistically meaningful difference (mean difference of −0.50 METs, $p = 0.067$, ES = −0.51 (95% CI—1.05–0.03)—large effect) was observed.

In addition, based on %HRmax of 77%, the mean HR was lower (mean difference of −7.38 bpm, $p = 0.10$, ES = −0.45 (95% CI—0.98–0.09)—large effect) but not statistically different, and maximum HR was statistically higher (mean difference of 14.55 bpm, $p = 0.009$, ES = 0.79 (95% CI 0.19–1.36)—large effect).

For RPE, there was an interaction between subjects using stimulants and whether they were in peak-week $F(1,53.80) = 6.21$, $p = 0.016$. Subjects taking stimulants displayed a higher RPE of 2.20 arbitrary units than subjects not reporting stimulant use ($b = 2.20$—large effect, $t(53.80) = 2.77$, $p = 0.008$). Also, subjects in peak-week displayed a higher RPE of 2.38 arbitrary units than subjects at other phases of preparation ($b = 2.38$—large effect, $t(53.80) = 3.01$, $p = 0.004$).

When comparing variables measured at one time point, subjects using stimulants displayed a higher mean HR (mean difference of 20.38 bpm, $p = 0.009$, ES = 1.50 (95% CI 0.31–2.64)—large effect size) and maximum heart HR (mean difference of 24.37 bpm, $p = 0.005$, ES = 1.63 (95% CI 0.42–2.79)—large effect size) compared to subjects not using stimulants. There was no difference for the other variables (Table 2).

When comparing variables measured at one time point, subjects at peak-week displayed a higher mean VO_2 (mean difference of 4.36 mL/kg/min^{-1}, $p = 0.008$, ES = 1.51 (95% CI 0.32–2.65)—large effect size) and METs (mean difference of 1.24 METs, $p = 0.008$, ES = 1.51 (95% CI 0.32–2.65)—large effect size) compared to subjects at other phases of preparation. There was no difference for the other variables, Table 3.

Discussion

The results of the present study confirm our initial hypotheses that bodybuilding athletes present a somewhat hard RPE and higher values of HR during bodybuilding posing. In addition, the posing sessions could be classified as

Table 1. Presentation time spent at bodybuilding posing, and rate of perceived exertion displayed by mean and standard deviation (SD).

	Time spent at posing	RPE
Set 1, min/s	02:84 ± 0.34	4.73 ± 2.08
Set 2, min/s	02:79 ± 0.38	5.80 ± 2.14
Set 3, min/s	02:77 ± 0.37	6.07 ± 2.08
Set 4, min/s	02:84 ± 0.46	6.47 ± 1.92
Total, min/s	11:25 ± 1.32	6.40 ± 1.63
Mean heart rate, bpm	135.07 ± 16.30	
Maximum heart rate, bpm	157.00 ± 18.50	
Estimated maximum heart rate, bpm	185.55 ± 5.47	
Total energy expenditure, kcal	99.47 ± 27.84	
VO_2, L/min	1.78 ± 0.37	
VO_2, mL/kg/min^{-1}	19.22 ± 3.45	
Ventilation per minute, L/min	61.16 ± 13.82	
Respiratory exchange ratio	0.92 ± 0.04	
METs	5.49 ± 0.98	

Note: rating of perceived exertion = RPE; beats per minute = bpm; mean oxygen consumption = VO_2; metabolic equivalents = METs.

Table 2. Comparison of the metabolic variables of the subjects under stimulants and under no use of stimulants displayed by mean and standard deviation (SD).

	Under use of stimulants ($n = 8$)	Under no use of stimulants ($n = 7$)	Hedges' g d ES and 95% CI	p
Mean heart rate, bpm	144.63 ± 12.48	124.14 ± 13.32	1.50 (0.31–2.64)	0.009*
Maximum heart rate, bpm	168.38 ± 12.98	144.00 ± 15.26	1.63 (0.42–2.79)	0.005*
Total energy expenditure, kcal	103.16 ± 28.64	95.24 ± 28.49	0.26 (−24.02–39.87)	0.60
VO_2, L/min	1.83 ± 0.38	1.74 ± 0.39	0.20 (−0.82–1.22)	0.69
VO_2, mL/kg/min^{-1}	20.26 ± 3.46	18.05 ± 3.30	0.61 (−0.44–1.64)	0.23
Ventilation per minute, L/min	62.95 ± 14.84	59.13 ± 5.07	0.25 (−0.77–1.27)	0.61
Respiratory exchange ratio	0.93 ± 0.04	0.92 ± 0.05	0.35 (−0.68–1.37)	0.49
METs	5.79 ± 0.99	5.16 ± 0.94	0.61 (−0.44–1.64)	0.23

Note: effect size = ES, mean oxygen consumption = VO_2; metabolic equivalents = METs; * Significant difference between groups, $p \leq 0.05$.

Table 3. Comparison of the metabolic variables of the subjects at the peak-week versus subjects at other phases of preparation displayed by mean and standard deviation (SD).

	At peak-week ($n = 8$)	Other phases ($n = 7$)	Hedges' g d ES and 95% CI	p
Mean heart rate, bpm	140.13 ± 15.22	129.29 ± 16.64	0.64 (−0.41–1.67)	0.21
Maximum heart rate, bpm	163.00 ± 20.42	150.14 ± 14.50	0.67 (−0.38–1.71)	0.19
Total energy expenditure, kcal	111.17 ± 31.53	86.10 ± 16.04	0.92 (−3.52–53.67)	0.08
VO_2, L/min	1.93 ± 0.44	1.62 ± 0.21	0.82 (−0.26–1.86)	0.12
VO_2, mL/kg/min^{-1}	21.26 ± 3.20	16.90 ± 2.02	1.51 (0.32–2.65)	0.008*
Ventilation per minute, L/min	65.71 ± 16.58	55.97 ± 8.13	0.69 (−0.37–1.72)	0.18
Respiratory exchange ratio	0.94 ± 0.04	0.90 ± 0.04	0.92 (−0.17–1.98)	0.08
METs	6.08 ± 0.91	4.83 ± 0.58	1.51 (0.32–2.65)	0.008*

Note: effect size = ES, mean oxygen consumption = VO_2; metabolic equivalents = METs; * Significant difference between groups, $p \leq 0.05$.

being of vigorous intensity using METs and maximum HR (e.g., 6.0 METs and 77% of HRmax) (Garber et al. 2011). Participants actively using stimulants displayed a higher and larger effect size for RPE, mean HR, and HRmax compared with athletes not reporting stimulant use. Athletes in peak-week displayed a higher and larger effect size for oxygen consumption and METs when compared to subjects at other phases of preparation.

The maximum HR during the posing training was significantly higher than predefined vigorous intensity measures used by scientific literature with a mean difference of 14.55 bpm of 77% HRmax (large effect size) (Garber et al. 2011), indicating strong contractions in the active muscle during posing stimulates sympathetic nervous activity (Seals 1989). The increased HR may have influenced the greater RPE of the athletes during posing training, as the more work

accumulated throughout the sets, the higher the RPE reported. There is a direct relationship between HR increases and RPE during exercise, and the magnitude of HR and RPE are equivalent when determining the intensity of effort (Scherr et al. 2013).

In addition, athletes who used stimulants like liothyronine (a thyroidal hormone T3 that is normally produced by the thyroid gland) (DRUGBANK 2023) and clenbuterol (a substituted phenylamino ethanol that has beta-2 adrenomimetic properties) (DRUGBANK 2023) showed higher mean and maximum HR responses during posing training when compared to those who reported not using stimulants. Stimulants contain substances that increase activity of beta-adrenergic receptors, resulting in tachycardia. However, the use of stimulants did not result in higher metabolic demands nor differences in body composition regardless of the increase in HR. Also, use of clenbuterol should be carefully analyzed as major adverse events reported in the scientific literature experienced by athletes who used clenbuterol was supraventricular tachycardia, atrial fibrillation, hypotension, chest pain, myocardial injury, myocarditis, myocardial ischemia, myocardial infarction, cardiomyopathy, hepatomegaly, hyperglycemia, and death (Kumari et al. 2023).

The effort required in posing training reflected the values found in the metabolic variables, with emphasis on oxygen consumption and MET spent, as well as total energy expenditure. Although not significant, such variables evaluated are slightly higher in the athletes who indicated the use of stimulants during the experiment as compared to those who stated that they did not use them. During caloric restriction, such as in the pre-competition period, any increase in calories expended may have an impact on tiredness, hungry, and general physiological state (Almeida et al. 2023).

During preparation, athletes report loss of lean mass and performance, while for bodybuilders, maintenance of lean mass is important for both performance and aesthetics (Kistler et al. 2014; Robinson et al. 2015). Athletes commonly use posing training as a strategy to increase caloric expenditure accompanied by the increasing volume of aerobic training and a high protein diet (Fagerberg 2018; Mitchell et al. 2018; Roberts et al. 2020). Moreover, our results reveal that even during the offseason period, the energetics of posing training should be considered for the calculation of a high caloric diet targeted to increase lean mass.

Bodybuilding posing is an essential practice for competitors, not primarily for energy expenditure and fat loss, but because it is a critical skill required for competition judging. Competitors pose in competitions to showcase muscle definition and symmetry, which are key judging criteria. Therefore, mastering the art of posing is crucial for competitive success. As highlighted in a previous work (Kind and Helms 2023), while posing is not primarily performed for its caloric burn, it does contribute to energy expenditure. Competitors must practice posing to acquire and refine this skill, ensuring they perform optimally during competitions. However, given the demands of their training regimens, which include RT, cardiovascular exercise, and strict dieting, competitors should be aware of the notable energy cost and physiological response associated with posing. This awareness is important for managing their overall energy balance and optimizing their preparation strategies for competition.

The strength of this study is that it is the first in the literature to address the energy expenditure, measurement of intensity expressed by METs and %HRmax and compared these values with the vigorous intensity standards proposed by the literature (Garber et al. 2011). It also contributes by showing that use of stimulants and phase of preparation affects HR response and RPE during posing. However, there are limitations in our study worth mentioning. Among the limitations, the small sample size of male participants can be highlighted, but justifiable, since competitive bodybuilders are rare due to the small number of athletes belonging to the sport. The absence of the exact doses and types of stimulants may be seen as a limitation. Also, each athlete has a different coach, supplement regimen, training, and diet program that may introduce bias when considering external validity of the results and interpretation of respiratory exchange ratio.

Although a respiratory exchange ratio of 1.0 can reflect an intense exercise (Herdy et al. 2016), participants did not follow recommendations before calorimetry as refrain from eating for at least 7 h, avoid caffeine and any stimulants for at least 4 h, and refrain from vigorous physical activity because some were in different competitive periods (Fullmer et al. 2015). In addition, as athletes were not familiarized with the OMNI scale as displayed by a previous study (Robertson et al. 2003), several biases can arise, potentially effecting the validity and reliability of RPE scores as interpretation bias, anchoring bias and central tendency bias. Despite the limitations, this quasi-experimental design can provide a more integrated picture of the existence evidence and can strengthen bodybuilding practice.

Conclusion

In conclusion, bodybuilding posing may be considered "vigorous exercise" when comparing to scientific literature using 6.0 METs and 77% of % HRmax. Also, reported use of stimulants as well as being in peak-week induce greater RPE, mean HR, maximum HR, total energy expenditure, and a higher VO_2 consumption.

Future recommendations

The results highlight the importance of posing training as an effective tool to increase total energy expenditure, HR, and RPE with a predominance of carbohydrates as energy fuel. This information can be used to improve diet calculations during different periods of preparation and to acknowledge the physiological stress of posing training.

Future research should incorporate simulated competitions to assess the physiological demands of posing in smaller, mid-size, and larger competitions with fewer or more competitors. This approach would provide a more comprehensive understanding of the energy cost and physiological responses, not only in intense posing practice sessions but also in actual competition scenarios. Additionally, studies should explore the posing requirements in other bodybuilding categories and between sexes to ensure the

literature accurately reflects the diverse demands of competitive bodybuilding. In addition, although no differences between groups were observed for BMI, it is crucial to consider this variable as an important covariate because higher BMI negatively impacts physical performance across various fitness domains. Specifically, a previous study indicates that individuals with higher BMI demonstrate lower endurance, decreased agility, and reduced relative strength (force produced per unit of body weight), despite having greater absolute strength (Leyk et al. 2006).

The use of stimulants should also be considered, especially during caloric restriction during pre-competitive phases. Moreover, the pre-competitive period increases the energy expenditure during posing training. This is the first study to show that bodybuilding is far from being just a "beauty show" of sculpted bodies. Bodybuilders are required to train technical aspects of posing and also present increased metabolic demand during this practice. This sport requires good physical condition and preparation to perform a posing presentation, such as in other sports.

Observing the athletes' mean and maximum HRs throughout the sets of posing reinforces the connection between increased muscular work and the response of the sympathetic nervous system. The correspondence between HR and RPE underlines the validity of this association, highlighting the importance of considering both cardiorespiratory responses and athletes' perception when strategically planning and evaluating posing training. The results also highlight the importance of understanding the metabolic variables involved in posing training as a strategy to preserve lean mass and to optimize caloric expenditure.

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Data availability

All authors agree with data availability. Considering the importance of open science practice, all raw data are freely available for review in the supplementary archive.

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Competing interests

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Ethics statement

Ethics approval and consent to participate. This study was approved by the Institutional Review Board (protocol number: 23163019.6.0000.0029). This study was conducted in accordance with the Declaration of Helsinki (World Medical 2013), and all athletes were informed about the study procedures and voluntarily signed an informed consent prior to enrollment.

Supplementary material

Supplementary data are available with the article at https://doi.org/10.1139/apnm-2024-0151.

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